

waves - practice problems

1. A pebble lands in a quiet pond and circular ripples spread outward. Why is the ripple better described with a wave model than a particle model?
2. Two students are 4.2 m apart and stretch a slinky between them. One student sends a pulse to the other end, and the pulse reflects back to the start in 2.8 s. What is the wave speed in the slinky?
3. Identify the medium for each wave: a sound wave from a speaker, a light wave from the Sun, a wave on a guitar string, and a traffic wave on a highway.
4. A small speaker plays a tone at 250 Hz. Each air molecule near the speaker oscillates back and forth many times each second. What is the period of the sound wave?

5. A bus line says a bus reaches the stop every 12 min. If the arrival pattern is treated like a repeating wave in time, what are the period and frequency of the arrivals?

6. A floating marker on a lake has an equilibrium height of 1.20 m above the lake bottom. As a wave passes, the marker reaches a crest at 1.70 m and a trough at 0.70 m. What is the amplitude of the wave, and what is the vertical distance from crest to trough?

7. A sound wave moves through air toward the right. A wave pulse on a rope also moves toward the right. How are the oscillations different?

8. A stadium wave takes 8.0 s for one bright jacket in the crowd to go from sitting, to standing, and back to sitting again. Treat that as one cycle. What is the frequency of the stadium wave at that seat?

9. A wave on a long spring has amplitude 0.30 m and period 0.50 s. The wave has been moving for 12 s. What is its wavelength?

10. A student shakes a rope harder and makes a larger amplitude wave. The rope tension and the rope itself stay the same. Should the wave travel faster just because the amplitude is larger?

11. A wave machine in a science museum makes water ripples with frequency 2.5 Hz. The crests are 0.80 m apart. What is the speed of the wave?

12. A low note from a large organ pipe has frequency 170 Hz. Use 340 m/s for the speed of sound in air. What is the wavelength of the sound in the room?

13. Two sound waves travel through the same room. One is a low note and the other is a high note. Which one has the shorter period, and why?

14. A microwave oven uses electromagnetic waves with wavelength 12 cm inside the oven cavity. Use 3.0×10^8 m/s for the wave speed. What is the frequency of the microwaves?

15. A guitar string is tightened to 45 N. The vibrating part of the string is 0.80 m long and has mass 0.012 kg. What is the speed of waves on the string?

16. A light wave enters glass from air and slows down. The frequency stays the same. What happens to the wavelength?

17. A 500 Hz sound from a small underwater speaker travels first through air and then through water. Use 340 m/s for sound in air and 1484 m/s for sound in water. What is the wavelength in each medium?

18. A laser beam hits a glass block. Some light bounces off the front surface and some light bends as it enters the glass. Which part is reflection, and which part is refraction?

19. A narrow light beam hits a flat mirror at an angle of incidence of 37° measured from the normal line. What is the angle of reflection?

20. A calm lake can show a clear image of the Moon, but a windy lake makes the Moon look broken into glittering patches. Which situation is more specular, and which is more diffuse?

21. A red laser has wavelength 6.0 mm in air in a scaled classroom model of refraction. The model uses index of refraction 1.00 for air and 1.33 for water. What wavelength should the model use for the laser in water?

22. A ray of light travels from air into glass. The angle in air is 30° from the normal. Use index of refraction 1.00 for air and 1.50 for glass. What is the angle in the glass?

23. In Snell's Law problems, why can you compare speed, wavelength, and index of refraction across two media, but not frequency?

24. A swimmer shines a waterproof flashlight upward from underwater toward the flat surface. Use index of refraction 1.33 for water and 1.00 for air. What is the critical angle for total internal reflection?